

Bioavailability of vitamin D from UV-B-irradiated button mushrooms in healthy adults deficient in serum 25-hydroxyvitamin D: a randomized controll...

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BACKGROUND/OBJECTIVES: Mushrooms contain very little or any vitamin D(2) but are abundant in ergosterol, which can be converted into vitamin D(2) by ultraviolet (UV) irradiation. Our objective was to investigate the bioavailability of vitamin D(2) from vitamin D(2)-enhanced mushrooms by UV-B in humans, and comparing it with a vitamin D(2) supplement. SUBJECTS/METHODS: Fresh mushrooms were irradiated with an UV-B dose of 1.5 J/cm(2), increasing vitamin D(2) content from <1 to 491 µg/100 g and made to an experimental soup. In this 5-week, single-blinded, randomized, placebo-controlled trial, 26 young subjects with serum 25-hydroxyvitamin D (25OHD) ≤ 50 nmol/l were randomly assigned into three groups ((a) mushroom, (b) supplement and (c) placebo). They received during winter (a) 28,000 IU (700 µg) vitamin D(2) via the experimental soup, or (b) 28,000 IU vitamin D(2) via a supplement or (c) placebo, respectively. RESULTS: After 2 weeks, serum 25OHD was significantly higher in the mushroom than in the placebo group (P=0.001). The serum 25OHD concentrations in the mushroom and supplement groups rose significantly and similarly over the study period by 3.9 nmol/l (95% confidence interval (95% CI): 2.9, 4.8) and by 4.7 nmol/l per week (95% CI: 3.8, 5.7), respectively. CONCLUSIONS: We are the first to demonstrate in humans that the bioavailability of vitamin D(2) from vitamin D(2)-enhanced button mushrooms via UV-B irradiation was effective in improving vitamin D status and not different to a vitamin D(2) supplement. This trial was registered at <http://germanctr.de> as DRKS00000195.